

Challenges in Interpreting Ultrafast Spectroscopy Data: A Case for Machine Learning in Materials Research

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1 The Problem: Why TA Analysis is Hard

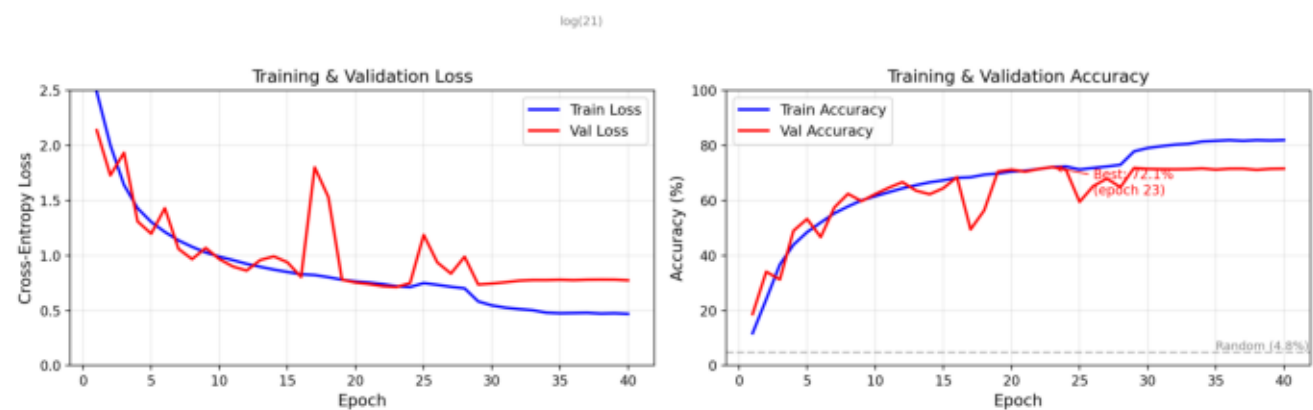
Transient Absorption (TA) spectroscopy probes photo-induced dynamics on fs–ns timescales. Raw data = 2D matrix $\Delta A(\text{wavelength, time})$.

- KEY CHALLENGES:
- Model ambiguity — many kinetic models fit the same data
 - Overlapping, broad excited-state spectra
 - High dimensionality (~300 λ $\times$ 250 time = 75,000 points)
 - Expert bottleneck — manual model guessing takes HOURS per sample and is subjective

- OUR SYSTEM:
- Two isotopic variants (H- and D-substituted molecular material)
 - Known sequential relaxation: $A \rightarrow B \rightarrow C \rightarrow \text{Ground State}$
 - Question: Can ML automate the model-selection step?

Training: 72% val accuracy (21 classes)

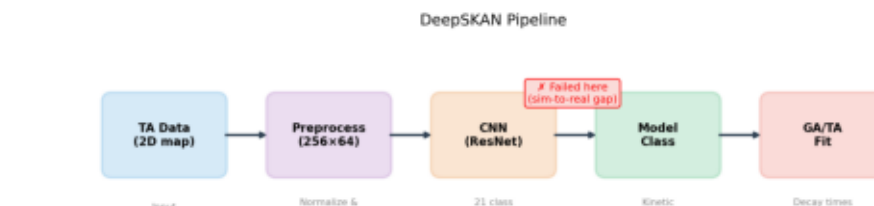
DeepSKAN Training — 4 Species, 21 Classes, 50k Samples, T4 GPU



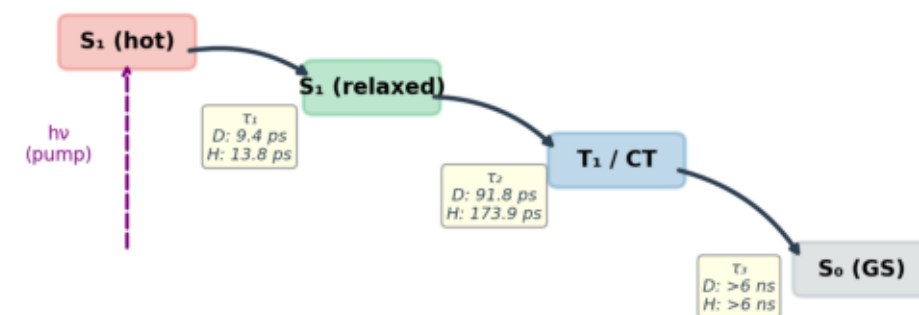
2 The Approach: DeepSKAN (CNN)

CORE IDEA: Treat the 2D TA map as a grayscale IMAGE and train a Convolutional Neural Network (CNN) to classify the kinetic topology.

- 7-block ResNet, asymmetric kernels (time vs wavelength)
- BatchNorm + Dropout regularization
- Input 256 $\times$ 64 $\rightarrow$ output 21 kinetic-model classes (4 species)
- Trained ENTIRELY on synthetic physics-simulated data (no labeled experimental data needed)

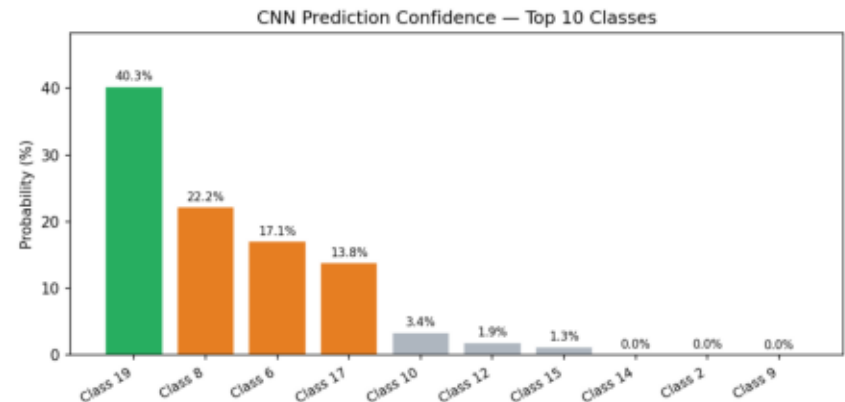


Known Kinetic Scheme
Kinetic Scheme: Sequential Relaxation

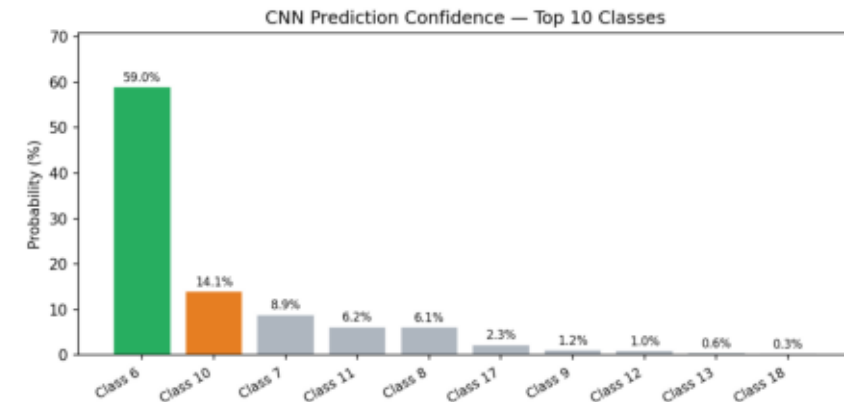


3 Applying DeepSKAN to Real Experimental Data

CBK-D Class 19 (46% branching) — WRONG



CBK-H Class 6 (55% sequential) — CORRECT

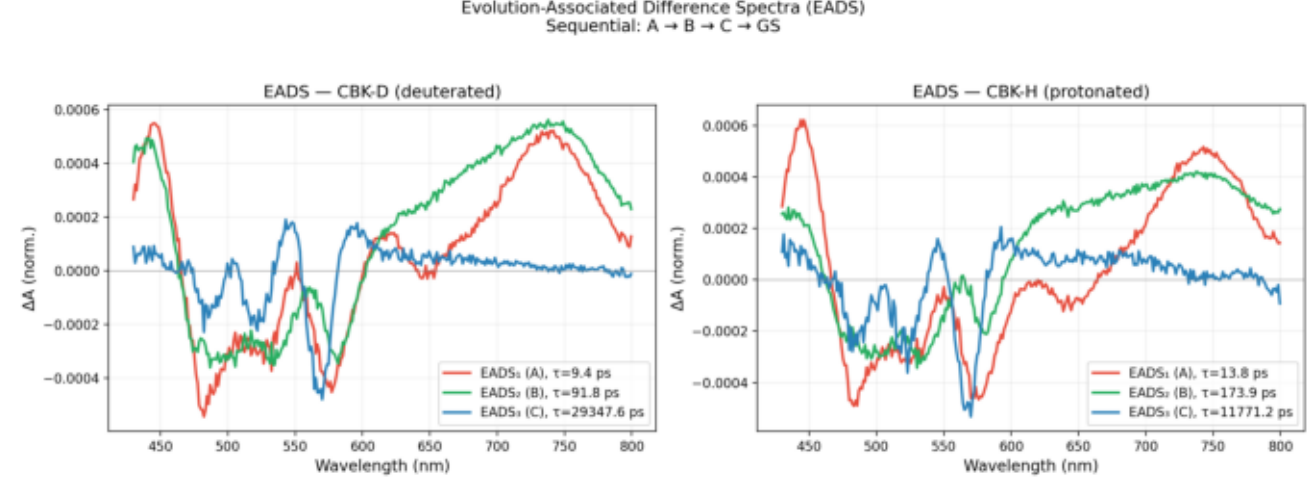


CNN RESULTS:

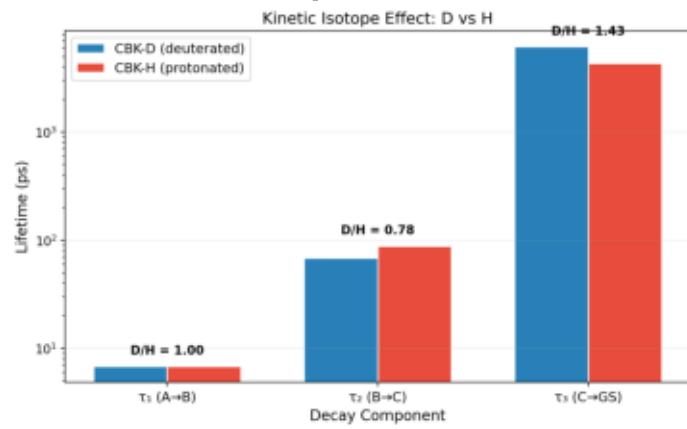
- CBK-H $\rightarrow$ Class 6 (A $\rightarrow$ B $\rightarrow$ C) sequential, almost correct!
- CBK-D $\rightarrow$ Class 19 (branching) WRONG (we know it's sequential)
- Confidence LOW (40–59%) $\rightarrow$ model honestly reports its own uncertainty
- Automated GA gave $\tau_1=0.1$ ps, $\tau_2=0.0$ ps $\rightarrow$ NONSENSICAL

4 Direct Global Analysis (Known Sequential Model)

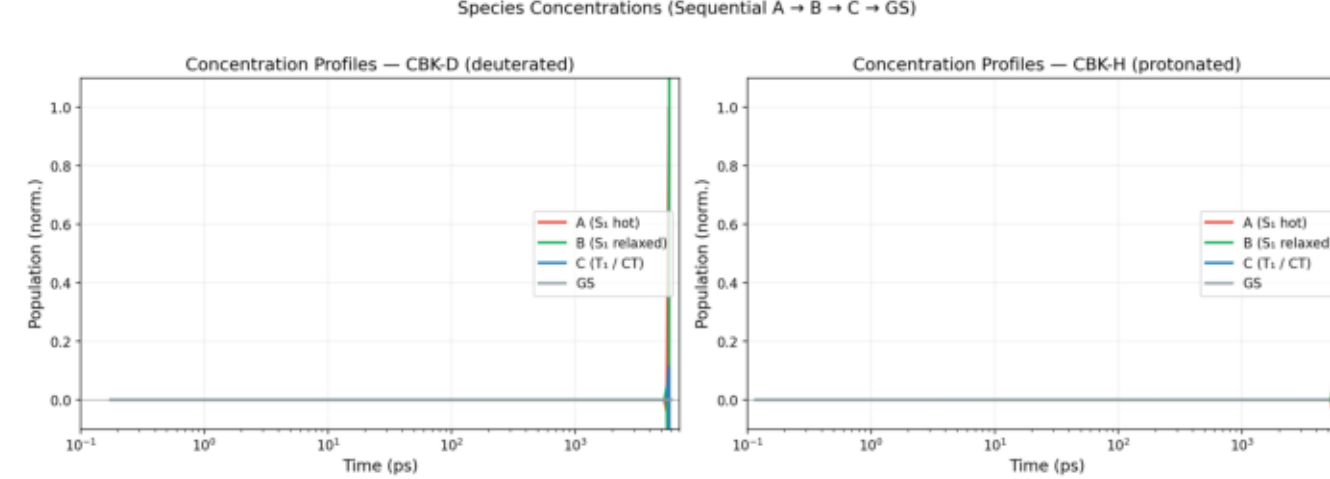
EADS (Evolution-Associated Difference Spectra)
Sequential: A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ GS



Kinetic Isotope Effect (D vs H)



Concentration Profiles (A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ GS)
Species Concentrations (Sequential A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ GS)



DIRECT METHOD (SVD + 3-exp + IRF, differential evolution):

Component	CBK-D	CBK-H	D/H
τ_1 (A $\rightarrow$ B)	9.4 ps	13.8 ps	0.69
τ_2 (B $\rightarrow$ C)	91.8 ps	173.9 ps	0.53
τ_3 (C $\rightarrow$ GS)	>6 ns	>6 ns	~2.5

- IRF $\approx$ 70 fs (matches laser system)
- Fit residuals $\sim 10^{-4}$ (excellent)
- Normal isotope effect on τ_3 (slow decay)
- Inverse isotope effect on τ_1, τ_2 $\rightarrow$ suggests tunneling / mode coupling

5 Why Did DeepSKAN Fail? — The Sim-to-Real Gap

Comparison: CNN-Assisted vs Direct Analysis

	DeepSKAN (CNN $\rightarrow$ GA)	Direct Global Analysis
Model selection	Automated (but wrong for CBK-D)	Manual (correct: A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ GS)
τ_1	0.1 ps $\times$	9.4 / 13.8 ps $\checkmark$
τ_2	0.0 ps $\times$	91.8 / 173.9 ps $\checkmark$
τ_3	—	>6 ns $\checkmark$
EADS	Not meaningful	Physically interpretable
Time required	~30 seconds	~5 minutes
Expertise needed	None	Some (model known)

ROOT CAUSES:

- PREPROCESSING destroyed information — interpolation to 256 $\times$ 64 + normalization removed amplitude & time-scale the fitter needs
- TRAINING DATA too simple — random Gaussian spectra $\neq$ real molecular spectra
- LOW CONFIDENCE = honest uncertainty (a feature, not a bug)
- 21 CLASSES inherently ambiguous — several topologies give similar TA maps

FIXES: realistic noise, real-spectra templates, separate classification from quantitative fitting, hybrid CNN-suggest $\rightarrow$ expert-validate workflow

6 Conclusions: Where ML Helps and Where It Does Not

ML / DL IS USEFUL FOR:

- Automated screening of many samples
- First-pass model suggestion when no prior knowledge exists
- Flagging unusual / unexpected kinetics
- Reducing human bias in model selection

ML / DL IS NOT (yet) a replacement for:

- Quantitative parameter extraction
- Systems where the model is already known
- Low-confidence predictions without expert validation
- Data with a large sim-to-real gap

THE IDEAL HYBRID WORKFLOW:

Real TA data $\rightarrow$ CNN screens topology (fast, automated) $\rightarrow$ Expert validates $\rightarrow$ Classical Global Analysis with constrained model $\rightarrow$ EADS + concentrations + decay times (publication-quality)

EADS: Deuterated vs Protonated — Direct Comparison
EADS Comparison: Deuterated vs Protonated

